

When an Alternative Becomes Possible: Lived Reorganization and Action-Option Generation Under a Common Perturbation History

Abstract

People can report that advice or information “finally made sense” despite earlier understanding; at other times, they can name an alternative while reporting that the situation feels unchanged. Existing work already separates supplied possibilities from generated options, links problem representation to option generation, and studies coherence among experiential, behavioral, and physiological responses under common elicitation. The present article asks a narrower identification question: under prospectively factorized histories of environmental evidence, can a report-derived representation of task-relevant relational/meaning organization and standardized first-production of a prelearned target action be predicted adequately by one preregistered shared response coordinate?

Hidden-Reroute varies cue diagnosticity and repetition without naming the target alternative. After matched evidence histories, separate trials sample immediate open reports, standardized option generation, explicit environmental belief, generic-state comparators, or reconstruction controls. Primary Φ excludes explicit option/action identity, and prior probe history is modeled because repeated measurement can alter later responses. Analysis first tests current-evidence sufficiency and a flexible shared-coordinate model class, then asks whether richer global response geometry is required. Because primary Φ may contain several frozen contrasts, global richer geometry is not by itself evidence of Φ/G separation. A prospectively frozen attribution gate first estimates the intervention-response structure required by Φ alone, then tests whether G is adequately contained within that structure or adds material held-out predictive geometry beyond it.

The proposal therefore tests a bounded measurement architecture, not a psychological mechanism. It does not infer simultaneous within-episode change, causal direction between Φ and G , latent candidacy, or ontological dimensionality.

Keywords: option generation; phenomenology; response coherence; affordances; decision making; experience sampling; reduced-rank modeling

1. When an Alternative Becomes Possible

People can encounter the same information repeatedly before anything recognizable changes. Advice can be heard, understood, remembered, and even endorsed while the action it recommends remains strangely external to the person receiving it. Someone may be able to explain why an alternative would make sense, or recommend it to another person, yet still fail to produce it as a live way forward when confronting the situation themselves. Then, sometimes, the words remain the same but the situation no longer seems organized in quite the same way. An alternative that previously sounded abstract, premature, threatening, disloyal, impossible, or simply “not for me” becomes something the person can imagine doing.

The reverse can also occur. A person may be able to state another action without reporting any corresponding change in how the problem is experienced. They can say what they could do while still feeling located inside the same problem. These possibilities motivate the central question of this paper: when an ongoing situation

begins to be experienced differently and when another action becomes generable from within that situation, do those changes constitute one transition, or can they respond differently to the same history of events?

The question is narrower than a theory of readiness, resistance, personality, psychotherapy, agency, or behavior change. It assumes no causal direction between lived reorganization and option generation and no necessary link to later action. It asks whether two response properties can be represented adequately by one preregistered shared response coordinate under controlled histories and, if the joint response vector requires richer geometry, whether G contributes intervention-response structure beyond that already required by Φ . Two variables should survive only if one cannot do the scientific work of both.

Ordinary language makes the problem visible but does not solve it. “I finally got it,” “I could see another way,” or “it finally felt possible” may refer to semantic understanding, changed problem meaning, generated action, preference, intention, attempt, or outcome. This paper isolates two relations: task-relevant lived organization (Φ) and standardized action-option generation (G). Figure 1 locates that comparison within the larger pathway without asserting a serial mechanism.

Because directly probing either response can change later cognition, the experiment does not measure Φ and G on the same critical trial. It estimates their intervention-response geometry across separately sampled branches under matched perturbation histories. The intuitive question may concern whether changes happen “at the same time,” but the assay does not observe same-episode simultaneity or temporal order.

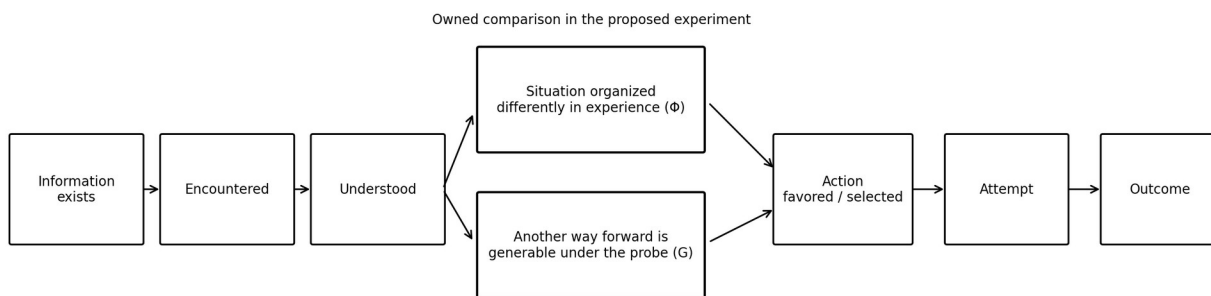


Figure 1. Different meanings of “it became possible.” The arrows are descriptive rather than a mandatory serial causal chain. The proposed experiment owns only the middle comparison between task-relevant lived organization (Φ) and standardized option generation (G). OpenAI ChatGPT (GPT-5.6) assisted diagram preparation; the author reviewed and verified the content.

2. Established Separations and the Remaining Gap

2.1 Generated options are not supplied possibilities

Option-generation research addresses decisions in which people must produce candidate actions before choosing. Johnson and Raab (2003) studied generation order and resulting choice; Kalis et al. (2013) argued for explicit study of the prechoice operation, and Kaiser et al. (2013) experimentally separated generation from subsequent choice. Retrieval-constrained valuation shows that retrieved alternatives constrain open-ended choice (Zhang et al., 2021), while generation and choice can rely on different value representations (Morris et al., 2021). Option generation can also involve ideation beyond memory retrieval (Del Missier et al., 2015).

These findings make several separations non-negotiable. A physically possible action is not thereby a generated option, and a generated option is not thereby the chosen action. A response to “what could you do?” also does not establish that the same option existed as a current candidate before the probe. The research target therefore contracts to directly measurable generability under a standardized operation: G records what is produced under the probe, not what was silently represented beforehand.

The human question can therefore be stated without inventing a hidden mechanism: was another way forward merely describable from the outside, or did the person become able to produce it from within the situation they were actually in?

2.2 Available affordances are not equally soliciting

Affordance research provides a neighboring precedent. An environment can offer multiple possibilities for action without those possibilities being equally relevant or inviting to a particular person in a particular situation. Dings (2018) argues that whether an affordance solicits action depends on its relevance to the agent's concerns and that solicitations can differ in experiential character. This is close to the motivating intuition here, but it should not be collapsed into the present variables. An affordance, a solicitation, an elicited option, and the lived organization of a problem are different scientific objects.

Jiang et al. (2024) provide a neighboring empirical example. Natural and built outdoor environments produced different patterns in affect and in the content of "I could ..." responses, while the number of generated action possibilities did not reliably differ. The study does not test the present hypothesis, but it shows that a common environmental manipulation need not move subjective and action-description measures in parallel.

2.3 Lived reorganization is not reducible to option generation

The experiential side is also strongly precedent. Insight research contains partly separable traditions centered on restructuring of the problem representation and on the phenomenological Aha experience. Wiley and Danek (2024) review these as related but non-identical targets. A new representation can occur without a dramatic Aha experience, and an Aha report does not by itself identify the cognitive route that produced the solution. That literature already blocks the inference that cognitive reorganization and phenomenological transition are interchangeable.

Affect and action repertoires have also been studied together. Fredrickson and Branigan (2005) manipulated emotion and measured subsequent thought-action repertoires. The present paper therefore does not claim that experience has never been linked to generated actions, or that response coherence is a new problem. It asks more narrowly whether this Φ/G pairing, under the declared perturbation and controls, is adequately predicted by one preregistered shared-coordinate model.

Direct experimental work also links problem representation or framing to option generation. Adelman, Gualtieri, and Stanford (1995) showed that different causal foci in an ill-structured problem changed the types of causal hypotheses and options participants generated and selected. Keller and Yang (2016) experimentally found that elements of foreign-policy problem representation, especially perceived threat, affected which options consciously occurred and which were not generated. Representation-to-generation influence is therefore PRECEDENTED rather than a Pull contribution. Neither study independently samples the present report-derived Φ channel and standardized G on separate branches or tests the $A \times N$ compression and attribution architecture developed here.

Emotion remains part of the human problem rather than a nuisance to remove mechanically. "I was less afraid" differs from "I was still afraid because leaving felt like abandoning them": the latter contains relational information about what the situation and action mean. Generic ratings of valence, arousal, difficulty, surprise, confidence, or effort are therefore lower-dimensional competitors, but Φ is not defined as residual variance after those ratings are removed.

2.4 The same information need not be the same psychological intervention

The motivating example of advice adds interpersonal structure that the controlled experiment intentionally removes. A recommendation can function as invitation, permission, demand, criticism, warning, reassurance,

or pressure despite similar propositional content. Freedom-threatening language increases reactance relative to lower-threat language (Li & Shi, 2026), while motivational-interviewing research distinguishes change and sustain talk subtypes whose relations to later outcomes differ (Magill et al., 2019).

These literatures are not evidence that Hidden-Reroute models advice, therapy, or relationship dynamics. They establish a narrower point needed for the human motivation of the paper: semantic sameness does not imply psychological sameness. Context, delivery, history, and autonomy can alter what an otherwise familiar recommendation does in the recipient.

2.5 Information can be understood without completing the next operation

Belief-updating theory supplies a conventional alternative to vague “resistance.” Sommer, Musolino, and Hemmer (2024) distinguish updating from evidence search, acceptance, hypothesis specification, integration, and reasoning: information can be present without every downstream operation being available or engaged. Retrieval may fail, search may not be extended, a relevant hypothesis may not be specified, evidence may not be integrated, or an action may remain low in accessibility or value. The present proposal adds no hidden barrier; it asks whether two measured response properties remain empirically interchangeable after such ordinary alternatives are taken seriously.

2.6 Immediate response is not persistence or realization

Immediate change, retained change, and later behavior are not one variable. Anglin et al. (2025) separated immediate and 24-hour belief response; Remmers et al. (2024) linked characteristics of real-world decisions to immediate mood and later implementation; and Mojzisch et al. (2020) manipulated option generation and measured regret later the same day. These studies do not test the present proposal. They establish that immediate response should not be used as a proxy for persistence or realization.

The primary experiment below therefore ends at immediate response organization. Persistence is retained as a separate extension rather than built into the central claim.

2.7 Response-system coherence already owns the generic compression question

Response-system coherence is already a mature same-level precedent. Mauss et al. (2005) measured within-person coherence among experience, facial behavior, and physiology; Evers et al. (2014) showed that coherence depends on which response systems are paired; Scherer (2009) treats synchronization among response components as a central dynamic question; and Mauss et al. (2024) review contemporary theory and measurement. Common elicitation, within-person coherence or dissociation, and response-class-dependent coherence are therefore precedented rather than contributions of this paper.

The overlap extends beyond emotion. Malo and Prié (2024) argue that affordance research should combine lived-experience and behavioral proxies rather than letting one substitute automatically for the other. Mixed first-person/behavioral measurement, response dissociation, option generation, and reduced-rank modeling are likewise not novelty claims here.

The remaining residue is narrower and OPEN. Response coherence/dissociation, option generation as a prechoice operation, problem representation or framing affecting generated options (Adelman et al., 1995; Keller & Yang, 2016), and reduced-rank/shared-index machinery are all PRECEDENTED. The residual composition combines a prelearned target action available before perturbation but never named by it; an immediate primary report representation excluding explicit action-option identity; standardized first-production of that target; E as a current-state sufficiency reduction; prospectively factorized $A \times N$ histories; condition-held-out tests of a preregistered shared-coordinate model against richer geometry; semantic, reconstruction, carryover, and contraction gates; and a nested attribution rule that separates Φ -internal

response structure from geometry contributed incrementally by G . A closer same-level precedent would contract the contribution to an application or synthesis.

2.8 The remaining identification problem

The component literatures already establish possibility versus generated option, generation versus choice, problem representation or framing effects on option generation (Adelman et al., 1995; Keller & Yang, 2016), restructuring versus Aha phenomenology, message content versus motivational effect, immediate versus delayed outcome, and the generic problem of within-person response-system coherence. The contribution cannot be another renaming of those precedented distinctions.

The surviving question is intervention-relative. Suppose one common history of events acts on an ongoing problem. We separately measure (a) how the current situation is organized in immediate task-relevant experience and (b) what action is generated under a standardized option-generation demand. Is this response pair adequately represented by the declared preregistered shared-coordinate model class? If not, does the richer joint response geometry arise entirely from structure required by Φ itself, or does G require intervention-response structure beyond the validated Φ -only response subspace?

A single endpoint dissociation is not enough. Thresholds, nonlinear readouts, delays, and noise can make outputs of one shared index look independent. A global preference for richer geometry is also not enough for a specifically Φ/G claim, because several frozen Φ contrasts may themselves require multiple intervention-response directions. The paper therefore separates global H_{2R} failure from attribution: Φ -only structure is established first, then a contained- G comparator asks whether G can be predicted within that structure, and an incremental- G comparator tests whether G adds material held-out predictive geometry beyond it. Rejection of H_{1D} remains model-class-relative and does not identify mechanisms or ontological susceptibilities.

3. Hypotheses and Formal Framework

3.1 Human-language hypotheses

The model hierarchy can be understood without mathematics.

Question 1: Is explicit understanding enough? Once two histories produce the same current judgment about what is happening, does the path by which the person arrived there still matter for either immediate lived organization or generated action?

Question 2: If history matters, can one preregistered shared response coordinate still suffice? Experience and option generation may change at different thresholds, scales, or noise levels while remaining different readouts of one shared index.

Question 3: Do different histories require richer response geometry, and if so, where is the extra geometry required? A global H_{2R} result can arise because Φ itself needs more than one intervention-response direction. The specifically Φ/G question is therefore whether G remains adequately predictable within the response subspace already required by Φ or instead adds material held-out predictive geometry beyond it.

3.2 Variables and analytical boundary

Let $P = (A, N)$ denote the controlled perturbation history, where A indexes cue diagnosticity or strength and N the number of informative observations. E is explicit current environmental judgment. Φ is a frozen report-derived representation of task-relevant lived organization; the report is evidence bearing on lived organization, not phenomenality itself. G is standardized option generation, with the primary outcome indicating whether the

prelearned target r^* is generated first. X denotes generic-state comparators and R delayed/reconstruction-control reports.

Because the design is repeated-measures, M records prior measurement/probe history before each critical observation: prior G , Φ , E , X , and R probes where applicable; total prior critical probes and order; and target/network-specific exposure. M is a carryover/design variable, not a proposed psychological stage or mechanism.

The analytical boundary is local to a learned goal-directed route-planning episode. The experiment does not identify psychotherapy, readiness, personality, or the causal origin of phenomenal experience. P is manipulated, so P -to-measure effects can be causal within the task; causal direction between Φ and G is not identified.

3.3 H_E: current-evidence sufficiency

The smallest account is current-evidence sufficiency. In schematic form,

$$H_E: P \rightarrow E \rightarrow \{\Phi, G\}. \quad (1)$$

The arrows express predictive sufficiency, not a serial psychological mechanism. H_E asks whether, once E and the declared measurement-history record M are known, the factorization of P adds material predictive information about Φ or G . Evidence-matched histories should therefore yield comparable response distributions. This is not mediation: if H_E predicts both response classes across matched histories, richer history variables are unnecessary for this query.

3.4 H_1D: one preregistered shared response coordinate

If history matters beyond E , one preregistered shared-coordinate model may still suffice. Let $h(P)$ be a preregistered feature representation of the perturbation condition, such as A , N , their interaction, and any prospectively fixed nonlinear basis terms needed to cover the calibrated transition region. For participant i and response contrast r , let η_{irc} denote the linear predictor or analogous latent response score for condition c .

$$\eta_{irc} = \alpha_{ir} + \beta_{ir}^T h(P_c). \quad (2)$$

H_{1D} constrains the response-specific coefficient vectors to one within-person perturbation direction:

$$\beta_{ir} = \lambda_{ir} d_i. \quad (3)$$

The shared direction d_i defines the model's common response coordinate; λ_{ir} allows response-specific scaling, and links, thresholds, category mappings, and noise may differ. The frozen Φ representation may contain several coded contrasts while G remains binary, provided all prespecified readouts depend on the same scalar index under H_{1D} . If Φ itself requires multiple perturbation indices, H_{1D} fails globally, but that is a measurement-architecture result and is not automatically a Φ -versus- G dissociation.

The rank-one H_{1D} restriction is therefore a global compression test. A failure caused entirely by multiple Φ contrasts cannot be attributed to G . The specifically Φ/G inference requires the separate nested attribution gate defined below rather than being read directly from the joint rank.

With a fixed feature basis $h(P)$, the matrix B_i whose rows are the β_{ir} vectors has rank at most one. This is conventional reduced-rank logic (Izenman, 1975), extended to categorical outcomes by reduced-rank vector generalized linear models (Yee & Hastie, 2003). The claim is relative to the declared function class: an inadequately flexible basis cannot establish a second response coordinate.

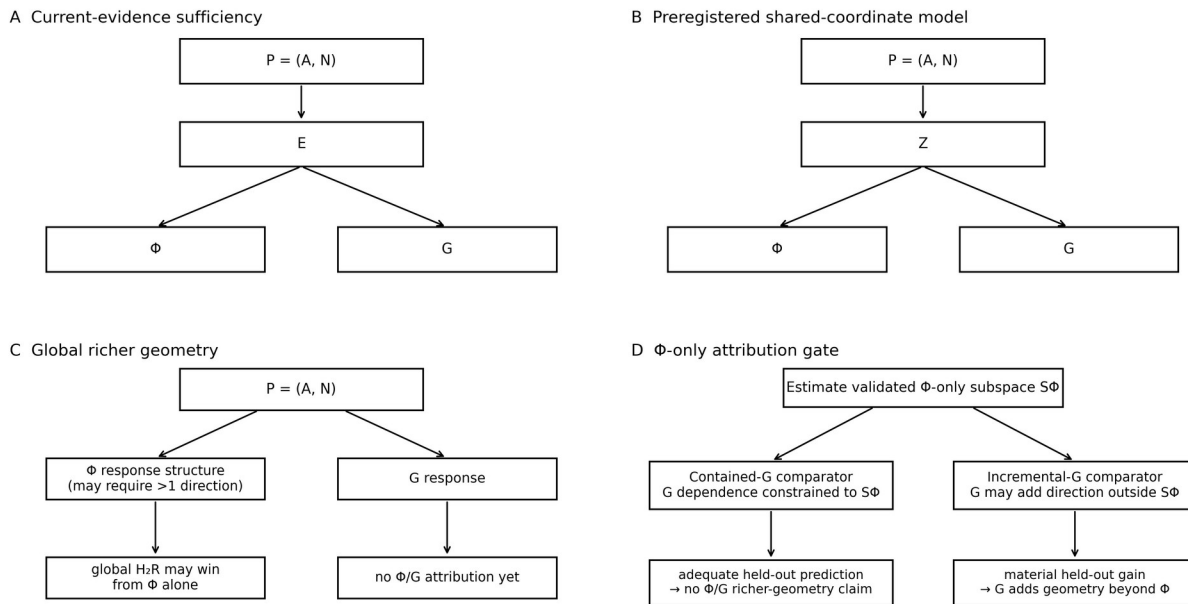
Before H_{2R} can survive, a flexible shared-index competitor must allow plausible nonlinear dependence on the calibrated $A \times N$ surface and response-specific readouts while retaining one shared scalar index.

Development simulations from that one-index family calibrate false-positive preference for H_2R under plausible nonlinearities, noise, and category structure. The feature basis, flexibility constraints, and calibration rule are frozen before confirmatory outcomes.

3.5 H_2R: a richer response architecture

H_2R relaxes the preregistered shared-coordinate restriction. Φ contrasts and G may depend on different directions through the same perturbation space while sharing causes, covariates, and interactions. In coefficient terms B_i may require rank greater than one; nonlinearly, one declared shared-index class may be insufficient. A global H_2R preference establishes only that the joint response vector requires richer geometry than H_1D. It does not localize that extra geometry to the relation between Φ and G.

The specifically Φ /G inference therefore has an additional attribution requirement. Let the validated participant-level intervention-response structure required by the frozen Φ representation alone define $S_{\Phi,i}$, or a preregistered nonlinear equivalent. $S_{\Phi,i}$ is a statistical response subspace, not a psychological object or stage. G is first fit with its own link, threshold, scaling, intercept, and noise while its perturbation-response dependence is constrained to $S_{\Phi,i}$. Only if that contained-G model is materially insufficient on held-out intervention conditions and an incremental-G model adds material predictive geometry beyond $S_{\Phi,i}$, while all existing gates and nonlinear safeguards pass, may the paper attribute richer geometry specifically to G beyond Φ . If Φ requires richer structure but G is adequately contained within $S_{\Phi,i}$, only Φ -internal structural dimensionality is licensed. None of these outcomes demonstrates separate anatomical systems, independent mechanisms, ontological susceptibilities, or causal direction between Φ and G. Figure 2 shows both the global model hierarchy and this attribution gate.



S_{Φ} is a preregistered statistical response subspace, not a psychological stage. M/carryover handling and the same validation gates apply throughout.

Figure 2. Nested explanatory alternatives and the Φ -only attribution gate. Panel A expresses current-evidence sufficiency; Panel B is the preregistered shared-coordinate model class; Panel C allows global richer response geometry but does not attribute that geometry to Φ versus G. Panel D first estimates the validated Φ -only response subspace S_{Φ} and then compares a contained-G model with an incremental-G model. A specifically Φ /G richer-geometry claim requires material held-out improvement from G geometry beyond S_{Φ} . S_{Φ} is a statistical response subspace, not a psychological stage. Measurement-history terms M are common carryover adjustments and are omitted from the diagram for clarity. OpenAI ChatGPT (GPT-5.6) assisted diagram preparation; the author reviewed and verified the content.

4. Study 1: Hidden-Reroute

4.1 Rationale and task

Hidden-Reroute instantiates the minimum structural problem. Participants learn matched route networks with recurring start states, destinations, constraints, and route families until they know the network and a target alternative r^* . On each critical trial an incumbent route r_0 is initially sensible, then the environment emits evidence about a latent disruption state $S \in \{\text{normal, disrupted}\}$. Under disruption, r^* becomes advantageous.

The environmental evidence never names r^* . It describes only the environment, preserving the distinction between generating a known alternative and choosing an option just supplied by the experimenter. Route identity, visual complexity, and assignment of r^* are counterbalanced or calibrated so the target is not privileged by naming, geometry, or practiced fluency.

4.2 Perturbation surface and evidence-matched histories

The history P has two controlled dimensions. A changes the diagnostic strength of each informative cue. N changes how many informative cues occur. Cue-window duration is held approximately constant across N by filling unused slots with neutral placeholders. This prevents repetition from being confounded trivially with elapsed exposure time.

A can be calibrated using the task's stipulated evidence model. If ℓ_k is the log-likelihood ratio contributed by cue k for disruption versus normal operation, cumulative normative evidence after N cues is $L_N = L_0 + \sum_k \ell_k$. Development work selects A levels that span the transition region and deliberately creates some histories with similar cumulative normative evidence but different factorization—for example, several weak cues versus one stronger cue. E then tests whether participants themselves reached comparable current environmental beliefs. The evidence model is a task-calibration device, not a theory of human belief updating.

A broad development grid can use four levels of A and four of N . The confirmatory set is chosen before confirmatory outcomes are inspected and must preserve three properties: broad variation in both dimensions, evidence-matched or near-matched histories with different factorizations, and enough two-dimensional intervention geometry to distinguish a common response direction from response-specific directions. If A and N collapse empirically to a single scalar before Φ or G are considered, the perturbation gate fails and the richer dimensional question is not identified.

4.3 Random measurement branching

After the common evidence history, each critical trial branches unpredictably to one measurement type. Participants do not know which measurement will be requested while the episode unfolds. This limits same-trial prompt contamination: G is not prompted on Φ trials and introspection is not inserted into trials later measured as G . It does not eliminate cumulative reactivity from earlier G , Φ , E , or X probes.

The principal branches are Φ (immediate open-ended experience), G (standardized option generation), E (explicit environmental belief), X (generic-state comparators), and R (delayed/reconstruction control). Exact branch probabilities and trial counts are development parameters rather than theoretical commitments. Figure 3 shows the current-trial branching architecture; the anti-carryover structure across trials is specified next.

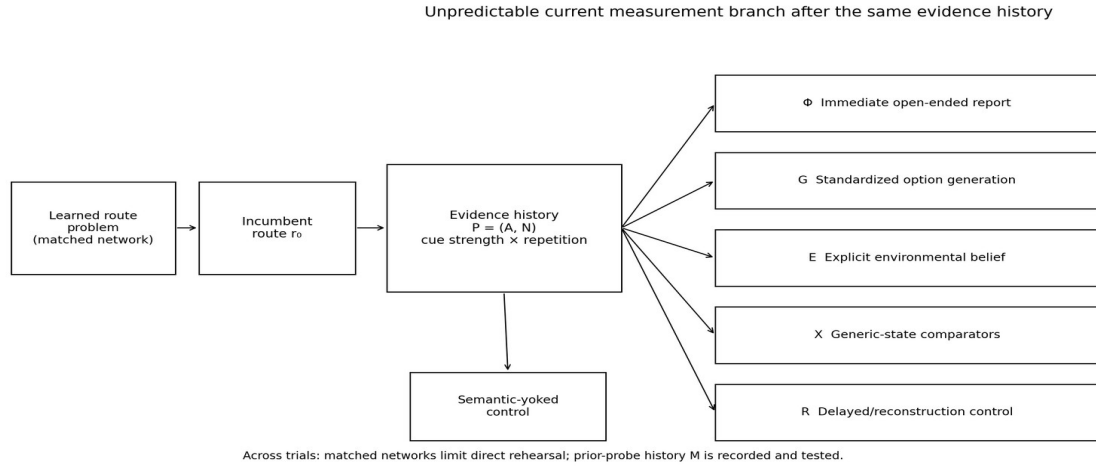


Figure 3. Hidden-Reroute architecture. After a learned route problem and incumbent route, the current evidence history $P = (A, N)$ branches unpredictably to one measurement class. Across trials, content-distinct isomorphic target networks limit direct rehearsal and prior-probe history M is recorded and tested for carryover. The semantic-yoked control preserves cue content while changing relevance to the active route problem. OpenAI ChatGPT (GPT-5.6) assisted diagram preparation; the author reviewed and verified the content.

4.4 Probe-history and anti-carryover architecture

Random branching limits same-trial contamination but does not reset the participant. Repeated experience-sampling can itself alter awareness or reporting (Eisele et al., 2023), and prior G probes may rehearse alternative search. The confirmatory design therefore uses multiple structurally matched but content-distinct route networks or target alternatives where feasible, counterbalances target assignment, and limits repeated exposure of the same target/network to the same probe family.

For each critical observation, M prospectively records prior G , Φ , E , X , and R probes where applicable; total prior critical probes and order; target/network-specific probe exposure; and whether the target family has previously received a G or Φ probe.

Carryover tests are preregistered. They include main effects of M , prespecified $P \times M$ diagnostics, and the same M adjustment across H_E , H_1D , and H_2R . The principal geometry must also survive a prospectively defined low-contamination sensitivity analysis based on first relevant probe exposure for a target/network and, where applicable, first G exposure within a target family. If repeated probing materially changes Φ or G and cannot be separated from P , the carryover gate fails; a separate validation block is added only if development shows the smaller design is inadequate.

4.5 Measuring Φ : lived organization without placing the theory in the prompt

On Φ trials, the task terminates unexpectedly and participants receive a neutral immediate probe:

“What, if anything, was most present in your mind immediately before this prompt?”

The prompt avoids asking whether perspective changed, another route was considered, the situation “felt different,” or an insight occurred. Those questions would import the theory into measurement. Descriptive Experience Sampling provides precedent for unpredictable probes aimed at immediately preceding experience (Hurlburt, 2017), while open-ended ESM can preserve context and relational information that fixed numerical items miss (Bringmann et al., 2026).

Participant language is the primary record before compression. Before development coding, primary Φ is bounded to task-relevant relational or meaning organization of the current problem and excludes explicit action-option identity, route proposals, “I could do X instead” content, implementation plans, and action sequencing already owned by G . Development coders, blind to A and N , derive only reproducible dimensions within that boundary; the codebook and primary contrasts are frozen before confirmatory analysis.

Excluded action-like content is retained in a secondary contamination channel coding explicit option/action mentions, route proposals, implementation plans, and action sequencing. It remains available descriptively and as a diagnostic of G -like leakage without defining primary Φ .

Before interpreting primary Φ , a preregistered validation asks whether it can be reconstructed within a declared material tolerance from same-report action-content indicators, participant/condition-level G from separate branches, X , and semantic cue paraphrase. If those simpler channels reproduce Φ without material loss, the stronger lived-organization claim contracts; if they exhaust the reliable signal, Φ does not survive as a distinct primary target.

A secondary semantic text representation may detect distributional changes missed by hand coding, but an opaque embedding cannot become the sole evidence for lived reorganization. Any semantic analysis interpreted as Φ must respect the same exclusion boundary. The primary claim requires an interpretable, prospectively frozen representation that survives blind application to the confirmatory sample.

4.6 Measuring G : standardized option generation

On G trials, participants receive a standardized neutral demand:

“What is one way you could proceed from here? Give the first thing that comes to mind.”

Participants respond through the route interface. The primary outcome, $G(1)$, records whether the prelearned target route r^* is the first generated action. On a prespecified subset, one additional search request can be made:

“What else could you do?”

This produces a secondary generation-depth measure, $G(2)$. If r^* appears only after the additional search demand, the licensed interpretation is that the option is generable under extended search but was not first-produced. The result does not establish that r^* was silently represented as a current candidate before the probe. That stronger candidacy question remains outside the assay. Treating first-produced order as an operational outcome is consistent with prior option-generation work linking generation order to resulting choice (Johnson & Raab, 2003), but $G(1)$ still does not identify latent candidacy.

4.7 Measuring E and X

E is measured on separate trials without asking about action. A representative item is a calibrated probability judgment such as “How likely is it that the network is currently disrupted?” The purpose is not to define all understanding by one rating. It is to estimate, for each participant and perturbation condition, the distribution of explicit current evidential judgment that can be matched across differently factorized histories. Because branching separates E from Φ and G on a given trial, H_E is a condition-level and participant-level sufficiency test rather than a same-trial mediation analysis.

X trials measure simpler state descriptions such as uncertainty, difficulty, effort, surprise, confidence, valence, and arousal. Their role is competitive rather than eliminative. A stronger Φ interpretation requires showing that the reproducible open-report organization is not fully reproduced by a generic state description. Φ is not defined as residual variance after every affective or metacognitive rating has been statistically removed.

4.8 Semantic-yoked and reconstruction controls

Immediate reports could simply paraphrase whatever cue content is salient. A semantic-yoked control therefore preserves the environmental signal sequence while changing whether the signaled feature matters to the active route problem. Cue semantics remain approximately constant; route relevance changes. If Φ merely reproduces cue wording, report organization should follow the cues similarly across the active and yoked problems. If task-relevant lived organization depends on the relation between the cue and the current problem, the response distributions may diverge.

A second control addresses retrospective reconstruction. Matched trials continue to completion before participants are asked what had been most present in mind. Delayed reports are compared with the immediate branch. If the predicted organization appears only after completion, retrospective reconstruction remains the more economical account. Immediate reports are not treated as transparent access to phenomenality; the timing manipulation only makes two routes to the same verbal endpoint less observationally equivalent.

5. Analysis, Identification Gates, and Model Comparison

5.1 Gates before dimensional inference

The dimensional comparison is not interpreted unless four prior gates are passed.

Phenomenology gate. Immediate reports must contain reproducible relational or meaning organization inside the predeclared primary Φ boundary and add material structure beyond action-option contamination indicators, condition-level G , generic X measures, and semantic cue paraphrase. If coding is unreliable or simpler channels reconstruct the confirmatory signal without material loss, the stronger Φ claim contracts.

Generation gate. The target route must be learned before critical trials but not supplied by the perturbation. If the environmental cue effectively tells participants to use r^* , or if the probe reduces to choosing the obviously dominant supplied route, G has not been identified as option generation in the intended sense.

Perturbation gate. A and N must preserve meaningful independent intervention geometry. If the calibrated histories are adequately summarized by one effective scalar before the response variables are examined, the experiment cannot adjudicate whether Φ and G require distinct response directions.

Carryover gate. The principal response-geometry conclusion must be stable after the declared M adjustment and in the prospectively defined low-contamination sensitivity analysis. If prior probing materially changes Φ or G , produces important $P \times M$ interactions, or leaves P inseparable from measurement history, the intended intervention-response geometry is not identified.

5.2 Hierarchical within-person model

The scientific target is within-person intervention-response geometry. Group averages are insufficient because population rank greater than one can be produced by mixing people who are each individually rank one but point in different directions. The primary model therefore pools information hierarchically while estimating participant-level response directions after the declared carryover handling.

Each frozen response uses an appropriate likelihood: Bernoulli for $G(1)$, multinomial or ordinal forms for categorical Φ contrasts when appropriate, and continuous likelihoods for numeric X or E measures. H_{1D} constrains participant-level P coefficients to the preregistered shared direction d_i ; global H_{2R} allows response-specific directions. Both include the same declared M main effects and $P \times M$ diagnostics. A global H_{2R} advantage is interpreted only as richer joint response geometry until the Φ -only attribution gate is applied. If carryover interactions prevent isolation of a stable P -response surface, the carryover gate fails rather

than being absorbed into H_{2R} . The reduced-rank constraint is conventional (Izenman, 1975; Yee & Hastie, 2003).

Regularization or hierarchical priors should stabilize participant-level slopes without forcing the rank result. Model checking includes condition-wise calibration, lapses, category sparsity, participant heterogeneity, and false-positive calibration under plausible one-index nonlinear data. The confirmatory hypotheses, feature basis, flexible shared-index competitor, M terms, response representation, and comparison criterion are fixed before confirmatory outcomes.

5.3 Attribution gate: Φ -only structure, contained G, and incremental G

Before a global H_{2R} result is used to make a specifically Φ/G claim, the frozen primary Φ representation is assessed alone using the same $h(P)$, development/confirmation separation, M treatment, semantic and reconstruction gates, regularization/calibration discipline, held-out-condition logic, and material predictive-benefit criterion used in the joint comparison. For participant i , let $B_{i,\Phi}$ denote the coefficient rows for the frozen Φ contrasts. The smallest validated Φ -only intervention-response subspace is denoted $S_{\Phi,i}$; in the linear representation:

$$S_{\Phi,i} = \text{rowspace}(B_{i,\Phi}). \quad (4)$$

The contained-G comparator then gives G its own link, threshold, scaling, intercept, and noise but constrains its perturbation-response coefficient to the already validated Φ -only subspace:

$$\text{Contained-G: } \beta_{iG} \in S_{\Phi,i}. \quad (5)$$

The incremental-G comparator relaxes only that containment, allowing G to use intervention-response structure not required by Φ alone:

$$\text{Incremental-G: } \beta_{iG} = s_{iG} + \delta_{iG}, \quad s_{iG} \in S_{\Phi,i}, \quad \delta_{iG} \notin S_{\Phi,i}. \quad (6)$$

A specifically Φ/G richer-geometry inference is licensed only if Φ -only structure is first established under the frozen rule, the contained-G model is materially insufficient on held-out intervention conditions, the incremental-G model yields material held-out predictive improvement, and the gain survives the same nonlinear shared-index, carryover, semantic, reconstruction, perturbation, and coding-fidelity safeguards. If Φ alone requires dimension greater than one but G is adequately contained within $S_{\Phi,i}$, the result is reported as richer Φ measurement geometry; no specifically Φ -versus- G richer-geometry conclusion is licensed. If $S_{\Phi,i}$ already spans the usable intervention-feature space, the incremental-G question is not identifiable under that basis. In nonlinear implementations, the same scientific rule is applied using the preregistered Φ -only function class and an equivalent containment comparison rather than treating the displayed linear subspace as ontological.

5.4 Hold out intervention histories, not merely trials

Random trial-level cross-validation would be a weak test because the model would repeatedly predict new observations from perturbation cells it has already seen. The stronger test holds out entire histories or cells. Models are fit on a subset of P conditions and required to predict responses under unseen perturbation conditions. Repeating this process across the surface tests whether a lower-dimensional architecture generalizes the intervention-response geometry rather than merely interpolating repeated observations from known cells.

Global H_{2R} survives only if the richer model provides a material gain under condition-level holdout relative to the preregistered flexible shared-coordinate competitors. Statistical detectability alone is not sufficient. For a specifically Φ/G richer-geometry claim, the contained-G comparator must also be materially insufficient and

the incremental-G comparator must improve held-out prediction after Φ -only structure is frozen. A gain confined to the Φ -only model is reported as Φ -internal structure, not Φ /G separation. The magnitude criterion should be preregistered in predictive units appropriate to the chosen likelihoods, with a smallest effect worth preserving set before confirmatory analysis.

5.5 Testing current-evidence sufficiency

H_E is evaluated first. Evidence-matched histories are identified prospectively and checked empirically using E . If histories with comparable E show no material held-out difference in Φ or G after the declared M adjustment, prior-history factorization is unnecessary for this query. If history still improves prediction after E and measurement history are accounted for, the shared-coordinate versus richer-geometry comparison becomes relevant.

5.6 Development, confirmatory sample, and sample size

This article does not present participant data and does not claim a preregistration-ready sample size. Several quantities needed for a defensible power analysis are intentionally left empirical: the base rate of first-generation of r^* , category frequencies in Φ , coding reliability, within-person slope variance, lapse rates, fatigue, and the useful A/N transition region.

Development precedes confirmation. Development data calibrate the route task and perturbation cells, derive and freeze primary Φ and its contamination channel, estimate nuisance parameters, quantify probe-history reactivity, determine the matched-network burden for low-contamination analysis, select the estimator, and freeze the Φ -only/contained-G/incremental-G attribution implementation. A simulation-based design analysis then sets confirmatory participant and trial counts for a preregistered minimum difference worth distinguishing among the declared model comparisons. Development data are not reused as confirmatory evidence.

5.7 Outcome logic stated before statistics

Table 1 states the strongest interpretation allowed for the main empirical patterns. Its purpose is to bind positive, null, carryover-failure, and contraction outcomes before model fitting.

Table 1. Outcome matrix. Interpretations are conditional on the relevant measurement, carryover, and perturbation gates.

Experimental pattern	Human reading	Strongest scientific interpretation
Comparable E predicts both responses across differently factorized histories	Once current evidence is matched, the path to it does not matter for these responses	H_E survives; richer history is unnecessary for this query
History matters, but Φ and G are predicted by one shared response coordinate	The measurements can differ in threshold or readout while remaining predictable from one common coordinate	H_1D survives; richer response geometry is unnecessary within the declared model class
Global H_2R beats H_1D, but Φ alone requires richer structure and G is contained in S_Φ	The multivariate response is richer, but G adds no direction beyond what Φ already requires	Report Φ -internal richer measurement geometry; no specifically Φ /G richer-geometry conclusion is licensed

Experimental pattern	Human reading	Strongest scientific interpretation
Φ -only structure is established and the contained-G model predicts held-out G adequately	G can have its own readout while remaining inside the intervention-response structure already required by Φ	G adds no incremental intervention-response direction beyond Φ under the tested model class
Contained-G is materially insufficient and incremental-G improves held-out prediction	The generated action responds along intervention geometry not required by Φ alone	G contributes intervention-response geometry beyond validated Φ -only structure under the tested boundary; no mechanism or causal independence is implied
Φ changes while G remains stable, or G changes while Φ remains stable	The two endpoints can dissociate descriptively under some histories	Descriptive dissociation only; the nested attribution test is still required for a Φ /G geometry claim
Primary Φ is reconstructed by action-option content, condition-level G, X, or semantic-yoked effects	The open reports add no distinct relational/meaning organization at the required level	Contract the Φ claim to the simpler measured channels
Predicted Φ organization appears only in delayed/reconstruction reports	The apparent organization may be produced after task completion rather than sampled immediately	Reconstruction control wins; do not interpret the pattern as immediate lived/problem organization
r^* appears only after “What else?”	The alternative is reachable under extended search but not first-produced	Retain generation-depth result only; do not infer pre-probe candidacy
Probe history materially changes responses and cannot be separated from P	The measurement process has become part of the effective intervention history	Carryover gate fails; the intended Φ /G response geometry is not identified

6. Failure, Contraction, and Inferential Boundaries

The proposal is designed to lose cleanly. If E suffices after M adjustment, richer historical structure is unnecessary. If the flexible H_1D class predicts both response classes, richer global geometry is unnecessary. If global H_2R wins but G is adequately contained within validated $S_{\Phi,i}$, the extra geometry is attributed only to Φ -internal measurement structure. A specifically Φ /G richer-geometry claim survives only when the contained-G comparator is materially insufficient and incremental G geometry improves held-out prediction under the frozen gates. If primary Φ is reconstructed by action-option content, G, X, cue paraphrase, or delayed reconstruction, its claim contracts. If probe history cannot be separated from P, the geometry is not identified. G contracts if the target is effectively supplied or generation adds nothing beyond ordinary route choice. The dimensional test also fails if A and N lack independent geometry, if $S_{\Phi,i}$ already exhausts the usable intervention-feature space for the incremental question, or if apparent higher rank exists only in group averages.

These outcomes are informative. A null result may show that the shared-coordinate class is sufficient, current evidence is sufficient, repeated measurement prevents identification, or a proposed measure did not earn its interpretation. The architecture then contracts to the smallest explanation that survives.

Even a successful global H₂R result remains nonmechanistic, and even a successful incremental-G attribution establishes only that G needs intervention-response structure beyond that required by Φ under the declared task, feature basis, and model class. Endpoint geometry cannot distinguish separate mechanisms from recurrent or shared processes with response-specific readouts, and it does not establish causal direction between Φ and G.

G is elicited option generation, not proof of latent pre-probe candidacy, causal participation in selection, selection itself, implementation, or realization. The experiment ends at susceptibility to generate an option under a standardized demand.

7. Persistence Is a Separate Extension

Study 1 asks what changes immediately. A later study can ask what remains changed. The same histories could be followed by a fixed delay and then re-sampled:

$$P \rightarrow \{\Phi_0, G_0\} \rightarrow \Delta t \rightarrow \{\Phi_1, G_1\}. \quad (7)$$

The extension could distinguish transient experiential change, delayed option generation, persistent generation with fading experiential reorganization, or stable coupling. It is not required for Study 1 because persistence is a different relation from immediate susceptibility.

8. Discussion

The phrase “another way became possible” is useful precisely because it is underspecified. Possibility may refer to the environment, semantic knowledge, an affordance, a solicitation, a generated option, a favored action, an intention, an attempt, or a realized outcome. The present framework does not replace that human language with one proprietary construct. It asks when two of its components can legitimately be compressed.

Most component machinery is established. Option-generation work separates generated alternatives from supplied choice sets and later choice (Johnson & Raab, 2003; Kaiser et al., 2013; Kalis et al., 2013; Morris et al., 2021; Zhang et al., 2021). Direct experiments also show that causal focus or problem representation can alter which options are generated (Adelman et al., 1995; Keller & Yang, 2016). Affordance, insight, affect, communication, belief-updating, and longitudinal research likewise establish neighboring distinctions used here (Anglin et al., 2025; Dings, 2018; Fredrickson & Branigan, 2005; Li & Shi, 2026; Malo & Prié, 2024; Sommer et al., 2024; Wiley & Danek, 2024). Pull therefore does not claim novelty for representation affecting option generation, framing effects on generated options, response coherence/dissociation, option generation as a prechoice operation, or reduced-rank/shared-index machinery.

Response-system coherence is also a mature same-level precedent: within-person coherence can be measured, can depend on the response systems compared, and need not imply one underlying mechanism (Evers et al., 2014; Mauss et al., 2005, 2024; Scherer, 2009). Common elicitation, coherence/dissociation, and generic common-versus-separable response organization are not contributions of this paper.

The contribution is narrower and compositional. Hidden-Reroute combines a prelearned target never named by the perturbation, a primary immediate report representation excluding option identity, standardized first-production of the target, E as a current-state sufficiency competitor, factorized $A \times N$ histories, probe-history control, condition-held-out shared-coordinate testing, semantic/reconstruction/contraction gates, and a nested

attribution rule that separates Φ -only structural dimensionality from geometry contributed incrementally by G . The burden is not merely whether the joint response vector is richer than H_1D ; it is whether G requires predictive intervention-response structure beyond that already required by the validated Φ representation.

If global H_2R materially outpredicts flexible shared-coordinate competitors after all gates pass, the first conclusion is only that the joint response vector requires richer geometry. If Φ alone requires that richness and G is adequately contained within S_Phi , no specifically Φ/G richer-geometry conclusion follows. Only when contained- G is materially insufficient and incremental- G improves held-out prediction may the manuscript state that G contributes intervention-response geometry beyond Φ under the tested boundary. Even then, the result would not establish two mechanisms or ontological susceptibilities, explain why people ignore advice, or define readiness.

Null outcomes are equally interpretable. H_1D may predict both measurements adequately, H_E may make prior-history factorization unnecessary, a measurement gate may fail, or carryover may prevent separating P from M . Each result contracts the architecture before a stronger dimensional claim is made.

Hidden-Reroute is deliberately artificial. Real advice and life decisions contain trust, attachment, fear, shame, loyalty, grief, identity, prior failed attempts, source credibility, autonomy, and social consequence. The route task strips most of this away. A positive result would establish identifiability in a controlled goal-directed system, not ecological equivalence to therapy, relationships, trauma, ADHD, or other emotionally significant decisions.

The human motivation still matters. “I knew that already. I just couldn’t see it then” can describe semantic understanding without lived reorganization, lived reorganization before another action is first-generated, or first-generation without corresponding reorganization. Ordinary conversation need not separate those possibilities; scientific explanation must if one measure is to stand in for another.

The remaining prior-art claim is deliberately limited. The search treats response coherence, affordance measurement, option generation, direct problem-representation/framing effects on option generation (Adelman et al., 1995; Keller & Yang, 2016), and reduced-rank/shared-index modeling as precedents. It did not identify the full composition tested here: primary lived/problem-organization reports separated from option content; first-production of a prelearned but uncued target; explicit current-evidence sufficiency; factorized $A \times N$ histories; probe-history control; held-out shared-coordinate versus richer-geometry tests; and a Φ -only versus contained- G versus incremental- G attribution rule with forced semantic, reconstruction, carryover, and contraction gates. Absence of an exact match is not proof of novelty, so the composition remains OPEN and would contract to application/synthesis if a closer same-level precedent is found.

9. Conclusion

A possibility can exist, be described by another person, and be understood without becoming the action a person produces when asked how to proceed. Problem representation and framing are already known to influence option generation, and existing science also supplies strong precedents for the component separations and response-system coherence used here. The unresolved issue is therefore not whether representation can affect generated options, but whether the two independently sampled response classes can be compressed by one declared intervention-relative response organization under the controlled reductions used here.

Hidden-Reroute samples task-relevant lived organization and standardized option generation on separate critical branches, measures current environmental understanding independently, and compares current-evidence sufficiency, a flexible preregistered shared-coordinate model, and global richer response geometry

under held-out intervention conditions. If global richer geometry is required, the attribution gate first establishes Φ -only structure, then asks whether G is adequately contained within that structure or instead adds material held-out predictive geometry beyond it. A specifically Φ/G richer-geometry conclusion is licensed only in the latter case. The experiment does not estimate same-episode co-occurrence, simultaneity, temporal order, causal direction, or mechanism.

Two variables should survive only if one cannot adequately do the scientific work of both. In human terms: can another way forward become easier to produce without the situation also becoming differently organized for the person under matched histories?

10. Data and ethics

No new data were created or analyzed in this study. Hidden-Reroute is a proposed experiment; any human-participant implementation would require prospective ethics review, informed consent, and preregistration of the confirmatory analysis.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used OpenAI ChatGPT interactively, under author direction, for literature identification and source organization, adversarial scientific critique and reduction, formalization, drafting and structural assistance, language editing, reference and submission-package organization, and figure/diagram assistance. The author originated the research questions and direction, made all substantive decisions, accepted or rejected arguments and sources, verified claims and citations, reviewed and revised the text, and takes full responsibility for the content of the published article. No empirical participant data or statistical research results were generated as findings by the AI tool.

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